

ThinkBench Segmentation Pipeline: Evolution from v1 to v3

Date: 2026-04-27

Scope: Technical history of `src/thinkbench/extract/segmenter.py` across three design generations.

Overview

The segmentation pipeline converts a raw chain-of-thought (CoT) trace into a list of **Thought Units** (TUs) — typed text segments — and a list of typed directed **Edges** between them. This is the foundational step: every downstream metric depends on what the segmenter produces.

The pipeline has gone through three substantively distinct designs, driven by two major design pivots:

1. **v1 → v2:** Remove the generative LLM from extraction (circular validity fix).
 2. **v2 → v3:** Replace surface-pattern edge detection with embedding-trajectory-based detection (model-agnostic robustness fix).
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v1 — Generative LLM Segmenter

Commit: `820e2d2` ("adding the pipeline code")

Lines: ~200

Returns: `list[ThoughtUnit]`

Architecture

A single LLM call via an `async def segment()` method on a `Segmenter` class. The model received the full raw CoT trace and was instructed to output a JSON array of thought units — each with its text, ID, and character offsets.

```
Raw CoT → LLM prompt → JSON parse → list[ThoughtUnit]
```

Fallback chain (when JSON parsing failed): 1. Regex split on headers and numbered list markers 2. Regex split on sentence boundaries

No edge types were assigned; the segmenter returned TUs only. Edge classification was presumably handled downstream or not at all at this stage.

Key properties

Property	Value
Boundary detection	LLM-generated (instructed via prompt rules)
Edge assignment	None
Model dependency	Requires LLM at extraction time
Reproducibility	Non-deterministic (temperature > 0)
Failure mode	Silent JSON parse failure → coarse fallback

Design rationale

At v1, ThinkBench was conceived as a fully LLM-driven pipeline. The reasoning was that a capable LLM could understand discourse structure better than hand-crafted rules. The segmenter prompt specified linguistic boundary cues (paragraph breaks, "Wait,", "Actually,", "Alternatively,", etc.) and asked the model to apply them.

Critical flaw

The LLM doing the segmentation is of the same family as the LLMs being analysed. This introduces **circular validity**: a GPT-4 segmenter may systematically segment GPT-4 traces differently from Qwen or DeepSeek traces — not because the reasoning differs, but because the segmenter recognises its own writing patterns. Results cannot be compared across models. This was the primary motivation for the v2 rewrite.

v2 — Non-Generative 3-Pass Pipeline

Commit: `fa67fce` ("v2 migration: non-generative extraction pipeline")

Lines: ~431

Returns: `(list[ThoughtUnit], list[Edge])`

Architecture

Three sequential passes, no LLM involved:

Raw CoT

- └ Pass 1 – Hard boundaries (regex cue phrases)
 - └ All 7 BoundaryClass types detected via pattern matching
 - └ Short spans merged into following span
- └ Pass 2 – Soft boundaries (TextTiling, MiniLM)
 - └ Cosine similarity between sentence windows
 - └ Local minima below τ = 30th percentile → NONE-class boundaries
- └ Pass 3 – NLI edge promotion (DeBERTa cross-encoder)
 - └ Local window (≤ 4 TUs): SEQ → SUPP / CONT / ELAB (threshold 0.75)
 - └ Non-local BACKTRACK pairs: → BACK (threshold 0.70)
 - └ Single batched predict() call per trace

Pass 1 — Hard boundaries

Seven `BoundaryClass` types were all detected by regex cue-phrase patterns applied to the first sentence of each candidate span:

BoundaryClass	Example phrases
BACKTRACK	"no wait", "actually", "hmm", "let me reconsider", "i was wrong"
BRANCH	"alternatively", "another approach", "what if instead", "or perhaps"
META	"let me step back", "i'm overcomplicating this", "let me re-read"
CONVERGENCE	"therefore", "to summarize", "so the key insight is", "the answer is"
ELABORATION	"specifically", "for example", "more precisely", "in other words"
CONTRAST	"however", "on the other hand", "in contrast", "but"
SUPPORT	"because", "since", "the reason is", "evidence for this"

Short spans (< 3 sentences AND < 50 tokens) were merged into the following span to avoid over-segmentation.

Pass 2 — Soft boundaries

Sentence embeddings (MiniLM-L6-v2) were computed for each NONE-class hard span. A sliding window (width = 3) computed cosine similarity at each sentence boundary. Local minima with prominence ≥ 0.15 and similarity below a per-trace τ (30th percentile of within-span sentence similarities) were promoted to NONE-class TU boundaries.

This was the first appearance of embedding-based detection, but only as a secondary "gap filler" for spans where cue phrases had not already assigned a boundary class.

Pass 3 — NLI edge promotion

For all SEQ-typed adjacent pairs (within window of 4), the DeBERTa cross-encoder tested three hypotheses: - SUPP: "The second passage provides evidence or justification for the first." - CONT: "The second passage contradicts or opposes the first." - ELAB: "The second passage is a more specific or detailed version of the first."

The best-scoring hypothesis above 0.75 replaced the SEQ edge. BACKTRACK spans > 4 apart were tested for a BACK hypothesis (threshold 0.70), linking each backtracking TU to the first prior TU above threshold.

All pairs were batched into a single `predict()` call per trace.

Critical flaw

The design worked well for models that use **explicit discourse markers** ("alternatively", "however", "because"). When tested on Qwen3.5-35B-A3B — which reasons in continuous, dense prose without explicit transitions — Pass 1 produced almost no BRANCH, ELABORATION, CONTRAST, or SUPPORT boundaries. The result: all TUs were classified SPC (default) or HYP (first TU), and the following metrics collapsed to 0.0 across all 90 traces:

- `branching_factor` (no BRCH edges → 0)
- `unique_perspective_count` (no RFR nodes → 0)
- `first_idea_diversity` (only 1 HYP node → 0)
- `self_reflection_rate` (no MET nodes → 0)
- `perspective_taking` (no RFR nodes → 0)
- `backtracking_rate` (no BACK edges → 0)

The 30-phrase cue-phrase lexicon was tuned for human-written think-aloud protocols and verbose reasoning models. It was fundamentally not portable to dense-prose LLMs.

v3 — Semantic Trajectory 5-Pass Pipeline

Current file: `src/thinkbench/extract/segmenter.py`

Lines: 556

Returns: `(list[ThoughtUnit], list[Edge], np.ndarray)`

Architecture

```
Raw CoT
├─ Pass 1 – Supplementary cue phrases (META / CONVERGENCE / BACKTRACK only)
│   3 out of 7 boundary types; BRANCH / ELABORATION / CONTRAST / SUPPORT
│   are no longer surface-detected – handled by Layer 3
├─ Pass 2 – TextTiling soft boundaries (unchanged from v2)
│   MiniLM sentence embeddings, window=3, prominence≥0.15
│    $\tau$  = 30th percentile of within-trace sentence similarities
├─ Layer 3 – Semantic trajectory (NEW – primary structural detection)
│   Per-TU cosine similarity to previous TU
│    $\tau_{\text{branch}}$  = 25th percentile  $\rightarrow \text{sim} < \tau_{\text{branch}} \Rightarrow \text{BRCH edge}$ 
│    $\tau_{\text{elab}}$  = 65th percentile  $\rightarrow \text{sim} \geq \tau_{\text{elab}} \Rightarrow \text{ELAB edge}$ 
│   Both thresholds are per-trace, data-derived
│   Cue-phrase overrides: CONVERGENCE  $\rightarrow$  SYNT, BACKTRACK  $\rightarrow$  BACK
├─ Layer 4 – Cross-segment analysis (NEW)
│   Synthesis:  $\cos(\text{emb}_j, \text{centroid}_s) > 0.50$  for  $\geq 2$  prior segments  $\rightarrow$  SYNT
│   Revision loop:  $\cos(\text{emb}_j, \text{emb}_i) > 0.65 + \text{gap drop} < 0.45 \rightarrow \text{BACK}$ 
│   (gap confirms genuine departure between i and j)
├─ Pass 5 – NLI refinement (demoted, only for untyped SEQ transitions)
│   Only runs on pairs not already typed by Layers 3/4
│   Threshold raised: 0.75  $\rightarrow$  0.78 (reduce near-universal SUPP over-triggering)
│   Window reduced: 4  $\rightarrow$  3 TUs
```

Design changes from v2

1. Cue phrases demoted from primary to supplementary signal

In v2, cue phrases were the primary mechanism for assigning all seven boundary types. If a phrase didn't match, the TU got NONE and was left to NLI.

In v3, cue phrases are **only used for three types** — META, CONVERGENCE, BACKTRACK — where the explicit surface marker carries unique semantic information beyond what embedding similarity can capture (e.g., "stepping back" signals meta-cognition; the embedding shift alone cannot distinguish this from a new hypothesis). The other four types (BRANCH, ELABORATION, CONTRAST, SUPPORT) are now assigned entirely from the embedding trajectory.

2. Adaptive per-trace thresholds replace fixed calibration

v2 used a single fixed threshold: τ = 30th percentile of within-span sentence similarities. This was applied globally to detect soft boundaries and was the same for every trace and every model.

v3 computes **two adaptive thresholds per trace** from the distribution of consecutive TU similarities:

- **τ_{branch} = 25th percentile:** The lowest-similarity transitions are topic shifts → BRCH.
- **τ_{elab} = 65th percentile:** The highest-similarity transitions are elaborations → ELAB.
- **Between τ_{branch} and τ_{elab} :** Neutral continuation → SEQ.

Because both thresholds are derived from the trace's own similarity distribution, the pipeline calibrates to the model's average transition distance. A verbose, repetitive model will have a higher absolute τ_{branch} than a terse model — but the relative classification (lowest 25% → BRCH) is consistent. This makes the pipeline **model-agnostic without retuning**.

3. Cross-segment analysis (Layer 4) — new structural layer

v2 had no mechanism for detecting cross-branch synthesis or revision loops beyond the local NLI window of 4 TUs. Long-range semantic returns (a TU 20 steps later returning to the topic of TU 3) were invisible.

Layer 4 adds two detectors operating over the full trace:

Synthesis detection: For each TU j , compute cosine similarity to the centroid of every prior topic segment. If j is highly similar (> 0.50) to ≥ 2 distinct prior segments, it is a synthesis point → SYNT edges from j to the representative TU of each bridged segment. The TU's `boundary_class` is updated to CONVERGENCE for the downstream classifier.

Revision loop detection: For each TU j , scan up to 25 positions back for a TU i (minimum gap = 4) in a different topic segment where $\cos(\text{emb}_j, \text{emb}_i) > 0.65$. A gap-drop criterion (minimum similarity in the gap between i and j must be < 0.45) confirms that the trace genuinely departed from topic i before returning — ruling out locally clustered sentences. → BACK edge from j to i , `boundary_class` updated to BACKTRACK.

4. NLI role reduced to gap-filling

In v2, NLI ran on **all** SEQ-typed local pairs within window 4. Because BRANCH, ELABORATION, CONTRAST, and SUPPORT were not assigned by cue phrases on dense-prose models, nearly every pair was SEQ, and the NLI ran on the full $O(n)$ adjacency list. The result was near-universal SUPP classification (DeBERTa finds weak entailment between almost any two consecutive sentences), giving `reasoning_density` ≈ 1.0 for all traces — an artifact rather than a signal.

In v3, NLI only runs on pairs that are **still untyped** (SEQ edge, no semantic edge from Layers 3/4). Since Layer 3 already assigns BRCH and ELAB to most transitions,

NLI sees a much smaller input. The threshold was also raised from 0.75 to 0.78 to further reduce SUPP over-triggering.

5. Embeddings returned to caller

v2 returned `(tus, edges)`. v3 returns `(tus, edges, embeddings)` — the $(n_tus \times 384)$ TU embedding matrix. This is passed directly to `classify_nodes()` in the classifier, enabling:

- **RFR vs. HYP disambiguation:** `cos(emb_i, emb_0)` range check for branch-receiving TUs.
- **Embedding-based MET detection:** TUs in the second half of the trace that return to the topic of `TU0` (`cos` $\geq$ 0.60) are classified as meta-reflection without requiring a cue phrase.

In v2, the classifier had no access to embeddings and could not make these distinctions.

Comparative Summary

Dimension	v1	v2	v3
Generative LLM required	Yes	No	No
Boundary detection mechanism	LLM prompt	Cue phrases (7 types)	Cue phrases (3 types) + embedding trajectory
Edge types from cue phrases	None (no edges)	BACK, BRCH, META→SEQ, SYNT, ELAB, CONT, SUPP	BACK, META→SEQ, SYNT only
Edge types from embeddings	None	Soft boundary (NONE class)	BRCH, ELAB (Layer 3); SYNT, BACK (Layer 4)
Threshold calibration	N/A	Fixed (30th pct)	Adaptive per-trace (25th + 65th pct)
Cross-branch synthesis detection	None	NLI (window 4)	Layer 4 centroid matching (full trace)
Revision loop detection	None	NLI BACK hypothesis (first match)	Layer 4 gap-drop criterion (best match)
NLI role	None	Primary edge promoter	Gap-filler for untyped SEQ pairs only
NLI threshold	N/A	0.75 (local), 0.70 (BACK)	0.78 (local only)
Embeddings re-turned	No	No	Yes (passed to classifier)
Token minimum to process	None	100	80
Short span threshold	None	<3 sent AND <50 tok	<2 sent AND <30 tok
Lines of code	~200	~431	556
Works on dense-prose models	Depends on LLM	No (cue-phrase dependent)	Yes (trajectory-based)

Impact on Metric Coverage

The table below compares the fraction of metrics that returned non-zero values across the 90-trace study, per pipeline version (v2 estimated from April 23 run; v3 from April 24 run):

Metric	v2 (estimated)	v3 (actual)
branching_factor	0.0 (no BRCH edges)	0.39 ± 0.12
unique_perspective_count	0.0 (no RFR nodes)	4.1 ± 2.8
first_idea_diversity	0.0 (<2 HYP nodes)	0.48 ± 0.21
self_reflection_rate	0.0 (no MET nodes)	0.07 ± 0.06
perspective_taking	0.0 (no RFR nodes)	0.23 ± 0.15
backtracking_rate	0.0 (no BACK edges)	0.12 ± 0.09
token_per_idea	= avg_tokens (bug)	198 ± 84 (rate)
domain_spread	1.0 (degenerate)	0.0 or 2.4 ± 1.1
reasoning_density	≈ 1.0 (NLI artifact)	0.68 ± 0.11

The v3 pipeline converted 8 degenerate metrics (returning a constant for all traces) into metrics with genuine within-trace and between-variant variance.

Remaining Limitations

- Pass 1 lexicon coverage:** The 30+ cue phrases for META/CONVERGENCE/BACKTRACK are still manually curated. Models that express these states without standard markers (e.g., through parenthetical asides rather than sentence-initial cues) will miss these boundary types.
- NLI disabled by default:** Pass 5 requires DeBERTa inference. In the current study it was disabled (`use_nli=False`) to isolate the embedding-trajectory signal. Re-enabling it should improve SUPP/CONT edge coverage for the final multi-model study.
- Single-newline traces:** The segmenter splits on double-newline (`\n\n`) for paragraph-level spans before sentence-level analysis. Traces formatted with single newlines between paragraphs (common in some model outputs) bypass Pass 1 splitting and enter Pass 2 as a single span, increasing NLI load.
- Fixed embedding model:** `all-MiniLM-L6-v2` (384 dimensions) is used for speed. The adaptive thresholds make the pipeline less sensitive to absolute similarity values, but a higher-capacity model (`gte-large-en-v1.5`, 1024 dimensions) would improve precision of BRCH/ELAB boundary placement, particularly for long traces.